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Shear-coupled migration governed by misorientation angles and atomistic structures in [1 1 0] symmetric tilt grain boundaries of face-centered-cubic Pt

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ABSTRACT

Shear-coupled grain boundary (GB) migration plays a crucial role in the deformation of polycrystalline materials. Although this phenomenon has been extensively investigated in [0 0 1] symmetric tilt GBs, other important GB systems remain relatively unexplored, hindering a comprehensive understanding of the underlying mechanisms. In this work, we systematically investigated shear-coupled migration in [1 1 0] symmetric tilt GBs of face-centered cubic Pt metal using large-scale molecular dynamics simulations by a high-efficient machine learning potential. For low-angle GBs, shear-coupled migration is found to proceed via collective intersecting dislocation glide, which generates steps on the second invariant plane. For GB approaching the $\Sigma 3(1\ 1\ 1)$ twin boundary, a twin-boundary-mediated migration mechanism is identified. Furthermore, the dependence of coupling mode on both the misorientation angle and the local GB configuration has been elucidated.

1. Introduction

With the reduction of material size to the nanoscale, the influence of grain boundary (GB) on material performance becomes increasingly dominant [1]. In particular, GB kinetics, which are directly related to the mechanical properties of materials, play a critical role in determining performance metrics such as strength, ductility and fracture toughness [2]. GB migration under applied shear stress has been widely observed and is known to significantly influence plastic deformation and grain growth in nanocrystalline metals. Consequently, shear-coupled GB migration has attracted considerable research interest and has been widely investigated [2–8].

A series of works has been devoted to elucidating the mechanism underlying shear-coupled GB migration. For symmetric tilt GBs in face-centered cubic (FCC) metals, Cahn et al. introduced the coupling factor (β) to characterize shear-coupled motion, and demonstrated that migration in selected [0 0 1] GBs follows two coupling modes, $\langle 1\ 0\ 0 \rangle$ and $\langle 1\ 1\ 0 \rangle$ modes, arising from the dislocation glides [4]. It is reported that the coupling mode is strongly dependent on the misorientation angle [4,9–11]. In particular, for [0 0 1] symmetric tilt GBs, the $\langle 1\ 0\ 0 \rangle$ mode dominates at the small tilt angles, whereas the $\langle 1\ 1\ 0 \rangle$ mode

prevails at larger tilt angles [4]. Furthermore, to describe shear-coupled migration under more complex conditions, several extended models have been proposed, including a geometrical model for general GBs [12], an extended method based on solutions of the Frank-Bilby equation [13], a GB mobility tensor [14] and a shear coupling strength tensor [15]. Notably, several studies have proposed the existence of second undistorted plane related to the coupling mode in addition to the first undistorted plane (GB plane) [10,12,13], which is similar to twin boundaries migration [16].

For low-angle GBs, shear-coupled migration proceeds via the glide of GB dislocations [4,13,17], with the coupling mode determined by the Burgers vector of the constituent dislocations [13]. Furthermore, a series of works found that the disconnection [18–20] composed of step and Burgers vector at the GB plays an important role in shear-coupled migration [4,5,21–36]. Under an applied shear stress, disconnections nucleate and propagate along the GB plane, thereby mediating GB motion [4,21]. This mechanism has been experimentally confirmed, for example, in the $\Sigma 11(1\ 1\ 3)$ GB in Au bicrystals, where both single-layer and double-layer disconnections have been observed [30]. The coupling factor can be quantitatively determined from the ratio of the magnitude of the Burgers vector (B) to the step height (H) of disconnection,

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enabling identification of the corresponding coupling mode [4].

Besides geometrical descriptions, the dynamical evolution of GB atomic structures during migration is critical for achieving atomistic insight into dislocation and disconnection processes [4,10,23,37,38]. Especially in $[0\ 0\ 1]$ symmetric tilt GBs, transformations of structural units (SUs) have been observed, resulting in disconnection nucleation [4]. Moreover, for the $\Sigma 9(2\ 2\ 1)$ GB, different shear directions parallel to the GB plane lead to distinct structural transformation pathways, resulting in markedly different migration behavior [23].

Due to the structural complexity and diversity of GBs, shear-coupled migration exhibits significant variability. In addition to the loading direction, variations in GB structural phases can alter the coupling mechanism, as demonstrated for the $\Sigma 5(2\ 1\ 0)$ GB [39]. Notably, previous studies have largely focused on relatively simple GB systems, such as low-angle GBs with single type of dislocation or GBs with single- or double layer disconnections [4,13,21,30]. However, GBs with more complex configurations may exhibit novel migration behaviors. For instance, kink formation during the disconnection motion [25] and dislocation interaction in GBs containing multiple defect types [24,40]. Therefore, systematic investigation of shear-coupled migration in structurally complex GBs is essential for developing a comprehensive mechanistic understanding.

FCC metals provide an important model system for studying shear-coupled GB migration. Among the various GB types, the shear-coupled migration in the $[0\ 0\ 1]$ symmetric tilt GBs has been well understood through the continuous development [4,9,10,41–43]. On the other hand, $[1\ 1\ 0]$ symmetric tilt GBs represent another significant class due to their high prevalence in polycrystalline materials [38,44,45]. Different with $[0\ 0\ 1]$ symmetric tilt GBs, previous study has identified a distinct $\langle 1\ 1\ 2 \rangle$ coupling mode in $[1\ 1\ 0]$ symmetric tilt GBs [9], particularly for misorientation angles in the range of 50° – 80° . This highlights the need for systematic investigations of shear-coupled migrations in $[1\ 1\ 0]$ symmetric tilt GBs to uncover new migration mechanisms.

Nowadays, owing to the rapid advancement of artificial intelligence,

the machine learning potentials (MLPs) have emerged as powerful tools for large-scale atomistic simulations, offering both high computational efficiency and near first-principles accuracy [46–48]. Despite their promise, applications of MLPs to GB remain relatively limited [48,49]. In this work, we develop a simulation framework for GB shear deformation based on large-scale molecular dynamics (MD) simulations enabled by a highly accurate neuroevolution potential (NEP). Benchmark tests demonstrate that this approach achieves high efficiency while maintaining accuracy comparable to first-principles calculations. Utilizing this framework, we systematically investigated shear-coupled migration in $[1\ 1\ 0]$ symmetric tilt GBs in FCC Pt. Several novel migration behaviors were revealed, including collective intersecting dislocation glide in low-angle GBs and twin-boundary-mediated migration mechanism. Furthermore, the influences of GB geometry and atomistic configuration on the coupling mode were elucidated. This work provides a comprehensive understanding of shear-coupled migration in $[1\ 1\ 0]$ symmetric tilt GBs, and establishes an efficient strategy for large-scale dynamic simulations of GBs with machine learning potentials.

2. Methodology

2.1. Construction of machine learning potential

To accurately and efficiently describe GBs in Pt, the NEP model was employed [50,51]. The reliability of this machine learning potential has been demonstrated in previous studies on inorganic halide perovskite [52,53], as well as metals and their alloys [46,54,55]. The training workflow of the NEP model is illustrated in Fig. 1(a). The first step involves the construction of an initial structure dataset. *Ab initio* molecular dynamics (AIMD) simulations were performed on several representative GB models at temperatures ranging from 300 K to 2400 K. To ensure adequate sampling of the configurational space, various $[1\ 1\ 0]$ tilt GB structures with lattice distortions and atom perturbations were generated. Besides the energetically lowest GB structures, the metastable

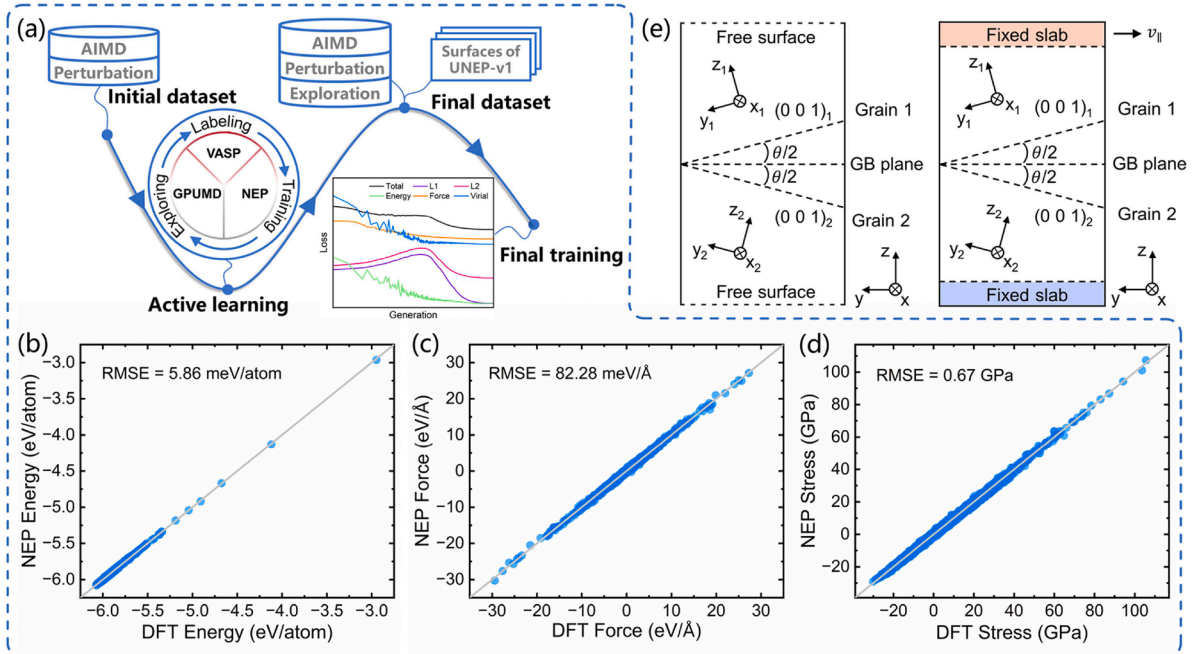


Fig. 1. (a) Workflow to construct the machine learning potential. (b)–(d) Comparison between NEP predictions and DFT calculations for (b) energy, (c) force and (d) stress, respectively. (e) Schematic diagrams of $[1\ 1\ 0]$ symmetric tilt GBs of Pt for stable GB structure searching (left) and shear loading simulation (right). The x -direction is along the tilt axis and the θ is the tilt angle around this direction. The $v_{||}$ is the applied velocity on the top fixed slab.

configurations such as the asymmetric tilt GBs shown in Fig. S1 of the Supplementary Material (SM) were considered as well to increase the structural diversity. In addition, bulk Pt structures with FCC, hexagonal close packed (HCP) and body-centered cubic (BCC) structures were perturbed to enrich the diversity of the dataset. Single-point energy calculations for these GBs and bulk configurations were performed based on the density functional theory (DFT). Additionally, since GB models with free surfaces were adopted in this work, surface structures of Pt from the UNEP-v1 dataset [46] were incorporated into the dataset. The NEP model was then trained based on this combined dataset.

As this study focus on GBs under shear deformation, it is essential to include sheared configurations in the training dataset. To this end, an active learning strategy was implemented. A series of GB models was generated by applying shear strains ranging from 0 to $\pm 15\%$ with increment of 0.5%. MD simulations were then performed on these models using a pre-trained NEP model. Those snapshot structures that were poorly predicted by the model were identified and added to the training dataset. Subsequently, the NEP model was retrained using the updated dataset. This iterative process was repeated until the calculated energies, forces and stresses from NEP showed satisfactory agreement with DFT results. More training details were shown in the SM. The testing results of energy, force and stress for well-trained NEP are shown in Figs. 1(b)–1(d). The mean square error (RMSE) values of testing results are sufficiently small. More tests including elastic constants, equation of state, phonon spectrum, general stacking fault energy (GSFE) curve, melting point, GB energies, and disconnection activation barrier (Table S2 and Fig. S3). The calculation results based on the NEP agree well with those from DFT calculations and experiments. These tests demonstrate high accuracy of trained NEP model. These tests demonstrate the high accuracy of trained NEP model.

2.2. Simulations of GBs under shear deformation

The shear response of $[1\ 1\ 0]$ symmetric tilt GBs in Pt at 300 K was simulated using the GPUMD software with the trained NEP model [51]. Here a single-boundary model was employed, with free boundary condition applied along the z -direction (perpendicular to the GB) and periodic boundary conditions along the x - and y -directions (parallel to the GB). To obtain stable GB structures, a rigid-body translation approach was adopted [56,57]. Two crystalline slabs with the required orientations were first constructed and combined using the CALYPSO and Atomsk packages (left panel of Fig. 1(e)) [58–61]. Then the relative rigid-body translations between the two grains were systematically explored to identify energetically favorable configurations. To avoid atom overlap, we removed the atoms whose interatomic distance is less than $0.75 \times \sqrt{2}a_0/2$ [56,57], where $\sqrt{2}a_0/2$ is the nearest-neighbor distance in a perfect FCC Pt crystal and $a_0 = 3.97$ Å. Geometry optimization was performed for all candidate structures, and the configuration with the lowest GB energy (γ_{GB}) was selected.

The γ_{GB} of GBs was calculated as:

$$\gamma_{GB} = \frac{\sum \varepsilon_i - N\varepsilon_{FCC}}{A} \quad (1)$$

where ε_i is the atomic energy of atom i , N is the number of selected atoms, ε_{FCC} is the atomic energy of a single Pt atom in bulk FCC Pt, and A is the GB area in a unit cell. In the selection of ε_i , to minimize the influence of top and bottom free surfaces, only atoms located within the central region ($0.25L_z$ to $0.75L_z$) were included in the calculation, where L_z is the length of lattice vector along the z -direction.

Following the identification of stable GB configurations, shear deformation simulations were carried out using MD. A velocity-driven loading scheme was adopted [39,43], in which atoms in the top and bottom regions of the bicrystal were fixed (Fig. 1(e), right panel). The thicknesses of these two regions were set to be at least twice the cutoff radius of the NEP model to eliminate surface effect. This GB model was

first equilibrated for 1 ns in the NVT ensemble at 300 K. Shear deformation was then applied by imposing a rigid-body displacement between the top and bottom fixed regions along the y -axis (v_{ij}) at a constant velocity of 1 m/s. The simulations were conducted for 1–3 ns with a timestep of 2 fs in NVT ensemble at 300 K. The resulting atomic trajectories were visualized using the Ovito software [62]. Structural evolution and defect dynamics were characterized using common neighbor analysis (CNA) method and the dislocation extraction algorithm (DXA) [63,64].

2.3. DFT calculations

DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP) [65]. The projector augmented wave (PAW) pseudopotentials method was employed to describe the electron-ion interaction [66]. The exchange-correlation effects were treated using the Perdew, Burke, and Ernzerhof (PBE) functional within the generalized gradient approximation [67]. A plane-wave kinetic energy cutoff of 400 eV was used and a Γ -centered k -point mesh with a spacing of 0.3 Å^{-1} was adopted.

3. Results

3.1. Structures of $[1\ 1\ 0]$ symmetric tilt GBs

It is known that GBs are commonly classified as low-angle and high-angle GBs based on the tilt angle. Previous studies have shown that migrations of low-angle GBs can be described in terms of dislocation glide, whereas the motion of high-angle GBs is usually interpreted using the disconnection nucleation [4,68–70]. Here both types of GBs were considered. As summarized in Table 1, four low-angle GBs with tilt angles approaching to 0° and 180° , and five high-angle GBs with tilt angles ranging from 50.48° to 109.47° , were selected. The corresponding atomistic structures were represented in Fig. 2.

It is interesting that, unlike previous low-angle GBs that typically consist of a single type of dislocation, the $\Sigma 243(1\ 1\ 22)$ and $\Sigma 163(1\ 1\ 18)$ GBs (Figs. 2(a) and (b)) are composed of two distinct $1/6\langle 1\ 1\ 2\rangle$ partial dislocations, which can be regarded as the nanoscale intrinsic stacking faults (ISFs). Besides, for the $\Sigma 33(4\ 4\ 1)$ and $\Sigma 51(5\ 5\ 1)$ GBs (Figs. 2(h) and (i)) with tilt angle close to 180° , two $1/2\langle 1\ 1\ 0\rangle$ dislocations are arranged alternatively along the GBs, and the dislocation cores consist of the E' SUs. The relatively high energy of these perfect dislocations contributes to the elevated GB energies for these two GBs. Figs. 2(c)–(g) show the high-angle GBs with SUs highlighted by gray lines. In particular, the $\Sigma 11(1\ 1\ 3)$ GB (Fig. 2(c)) consists of a periodic chain of C SU, consistent with previous reports for other FCC metals [30,44,68]. The

Table 1

Summary of stable GB information, including coincidence site lattice (CSL) GB notation, tilt angle, GB energy and description of periodic GB structure based on dislocations or SUs.

GB	Tilt angle ($^\circ$)	γ_{GB} (mJ/m 2)	Structural description
$\Sigma 243(1\ 1\ 22)$	7.36	554.5	$1/6\langle 1\ 1\ 2\rangle$ dislocation array
$\Sigma 163(1\ 1\ 18)$	8.98	621.0	$1/6\langle 1\ 1\ 2\rangle$ dislocation array
$\Sigma 11(1\ 1\ 3)$	50.48	335.8	CC
$\Sigma 3(1\ 1\ 2)$	70.53	580.5	EE
$\Sigma 17(3\ 3\ 4)$	93.37	709.9	DDEDDE
$\Sigma 57(4\ 4\ 5)$	97.05	663.8	DDDEDDE
$\Sigma 3(1\ 1\ 1)$	109.47	133.9	DD
$\Sigma 33(4\ 4\ 1)$	159.95	793.4	$1/2\langle 1\ 1\ 0\rangle$ dislocation array or A'E'A'E'
$\Sigma 51(5\ 5\ 1)$	163.90	726.1	$1/2\langle 1\ 1\ 0\rangle$ dislocation array or A'E'A'E'

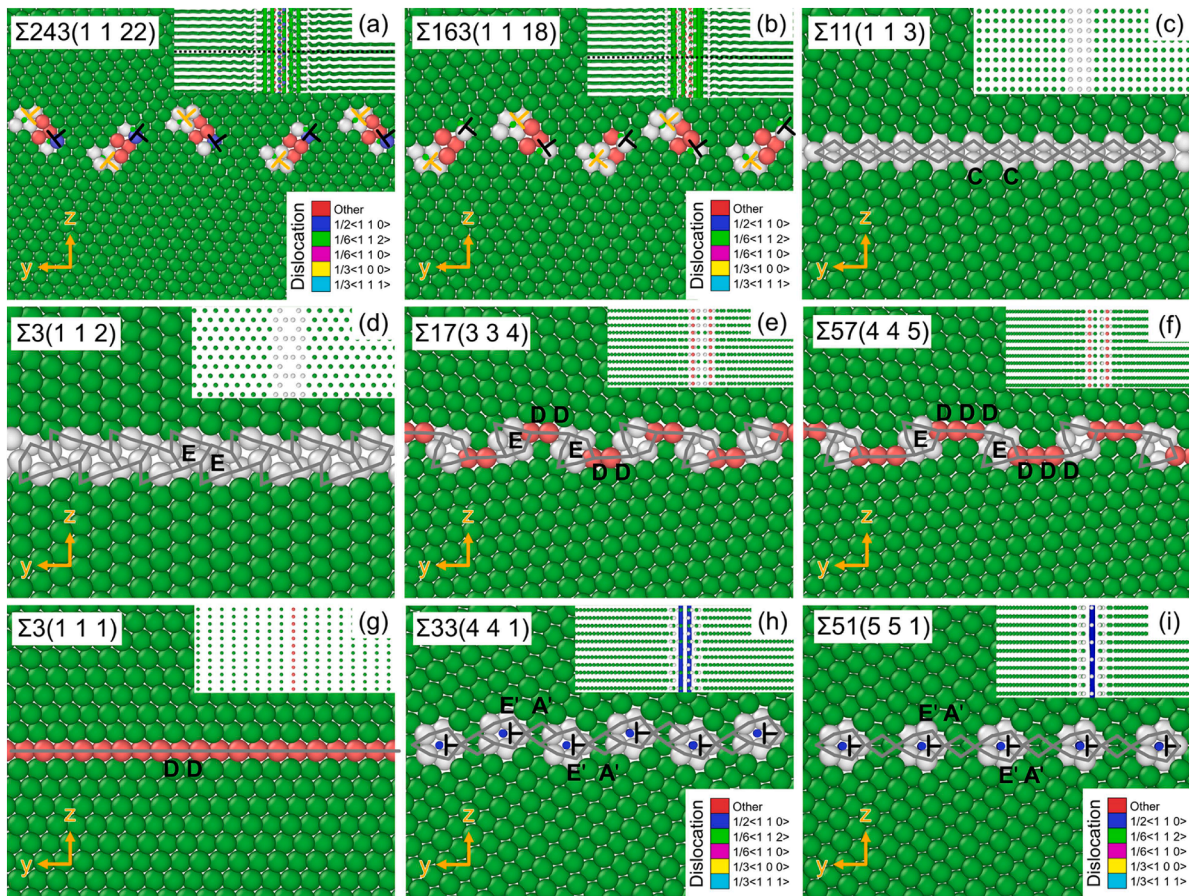


Fig. 2. Optimized structures of $[1\ 1\ 0]$ symmetric tilt GBs for Pt. (a) $\Sigma 243(1\ 1\ 22)$, (b) $\Sigma 163(1\ 1\ 18)$, (c) $\Sigma 11(1\ 1\ 3)$, (d) $\Sigma 3(1\ 1\ 2)$, (e) $\Sigma 17(3\ 3\ 4)$, (f) $\Sigma 57(4\ 4\ 5)$, (g) $\Sigma 3(1\ 1\ 1)$, (h) $\Sigma 33(4\ 4\ 1)$ and (i) $\Sigma 51(5\ 5\ 1)$. The atomic color represents structural type with green for FCC, red for HCP, blue for BCC and white for others. The GB SUs are highlighted with gray lines. The distinguishable dislocations are identified by DXA method and the color of dislocation line signs the dislocation type. The orange and black “ $\perp$ ” shapes represent the mixed and edge dislocations, respectively. The insets show the side views of GB structures.

$\Sigma 3(1\ 1\ 2)$ GB is composed of E SU, while the $\Sigma 3(1\ 1\ 1)$ GB corresponds to a coherent twin boundary characterized by D SU. Moreover, the $\Sigma 17(3\ 3\ 4)$ and $\Sigma 57(4\ 4\ 5)$ GBs exhibit mixed configurations consisting of D and E SUs.

3.2. Shear-coupled migrations of $[1\ 1\ 0]$ symmetric tilt GBs

MD simulations of GBs with shear deformation revealed shear-coupled migration behavior for all investigated models. The temporal

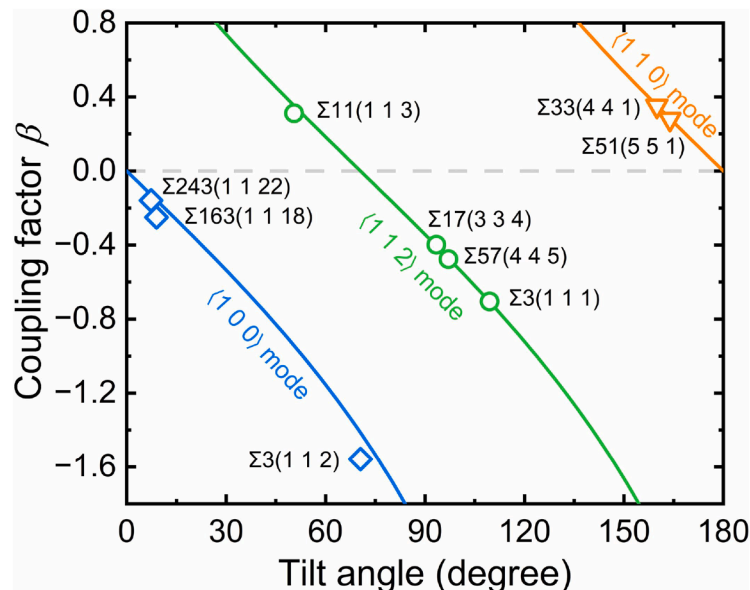


Fig. 3. The coupling factor β obtained by MD simulations as a function of the tilt angle. The lines represent three theoretical branches of shear coupling modes.

evolution of GB movements and shear stresses were presented in Fig. S4. The coupling factor, β , defined as $v_{\parallel}/v_{\perp}$, where $v_{\parallel}$ and $v_{\perp}$ denote the tangential grain translation velocity and the normal GB migration velocity, respectively, was calculated [39]. The dependence of β on the tilt angle was shown in Fig. 3. This result shows that the calculated β values agree well with the coupling modes reported in previous studies [9], thereby validating the present simulations. In contrast to $[0\ 0\ 1]$ symmetric tilt GBs, a new $\langle 1\ 1\ 2 \rangle$ coupling mode is observed, which dominates the high-angle GBs except for the $\Sigma 3(1\ 1\ 2)$ GB. For low-angle GBs, those with tilt angles approaching 0° follow the $\langle 1\ 0\ 0 \rangle$ mode, whereas those near 180° exhibit the $\langle 1\ 1\ 0 \rangle$ mode.

3.3. $\langle 1\ 0\ 0 \rangle$ mode

$$\frac{1}{6}[\bar{2}\ \bar{1}\ 1] + \frac{1}{6}[\bar{1}\ 1\ 2] + \frac{1}{6}[2\ 1\ 1] + \frac{1}{6}[\bar{1}\ \bar{1}\ 2] \rightarrow \frac{1}{6}[\bar{1}\ 1\ 0] + \frac{1}{6}[\bar{1}\ 1\ 2] + 2 \times \frac{1}{6}[\bar{1}\ \bar{1}\ 2]. \quad (2)$$

There are three GBs including $\Sigma 163(1\ 1\ 18)$, $\Sigma 243(1\ 1\ 22)$ and $\Sigma 3(1\ 1\ 2)$ GBs exhibit the $\langle 1\ 0\ 0 \rangle$ coupling mode, with the coupling factor given by $\beta_{\langle 1\ 0\ 0 \rangle} = -2\tan(\theta/2)$. Despite sharing the same coupling mode, their migration mechanisms differ significantly between low- and high-angle GBs. For the low-angle $\Sigma 163(1\ 1\ 18)$ GB, the GB dislocation and migration process under shear loading were illustrated in Figs. 4 and S5. Four individual $1/6\ \langle 1\ 1\ 2 \rangle$ dislocations superimpose, leading to the

formation of an effective $\langle 1\ 0\ 0 \rangle$ Burgers vector, which governs the observed coupling mode. Interestingly, contrary to the previously reported $[0\ 0\ 1]$ symmetric tilt GBs that typically involve a single dislocation type [4,71], we found that the shear-coupled migration in this GB proceeds via the collective glide of multiple $1/6\ \langle 1\ 1\ 2 \rangle$ dislocations. Under applied shear, these dislocations glide along their respective slip planes. Furthermore, dislocation reactions occur when two dislocations meet at the intersection region (marked by red square in Fig. 4(b)), producing a $1/6\ \langle 1\ 1\ 0 \rangle$ stair-rod dislocation (Fig. 4(d)). Figs. 4(c) and (d) show the process of forming a stair-rod dislocation through the reaction of

This newly formed defect acts as a sessile dislocation, impeding further glide and causing temporary GB thickening (Fig. 4(e)). With continued shear loading, the stair-rod dislocation partially reverts to a glissile dislocation and moves along the ISF (Fig. 4(e)). Then the glissile and sessile parts alternate, resulting in a size reduction of the ISF (Fig. 4(f)). Under sustained shear loading, the stair-rod dislocation moves into the intersection region with another dislocation pair, and reacts with

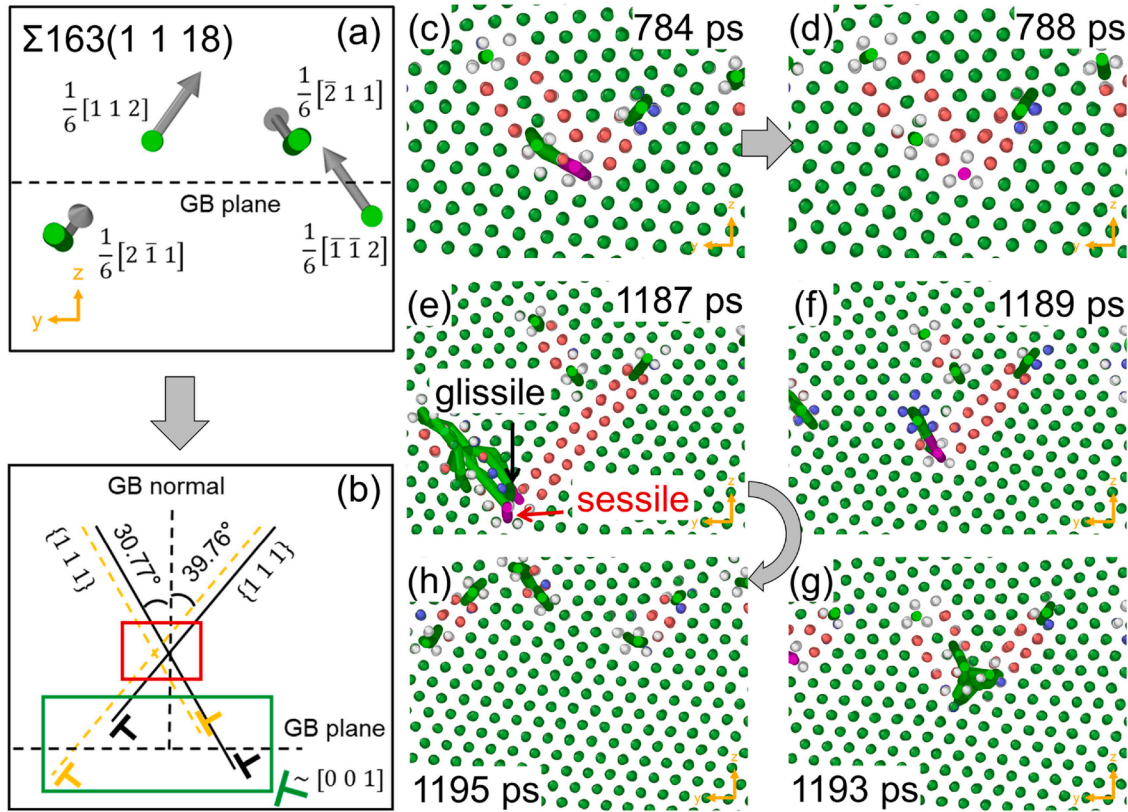


Fig. 4. (a) Typical dislocation structure in $\Sigma 163(1\ 1\ 18)$ GB at 0 K. (b) Schematic diagram of dislocation structure in $\Sigma 163(1\ 1\ 18)$ GB. The orange and black “ $\perp$ ” shapes represent the mixed and edge dislocations, respectively. The red square highlights the intersection region of slip planes of the two dislocation pairs. The green “ $\perp$ ” shape shows the total Burgers vector looped by the green square. (c)–(h) show the evolution of GB dislocations under shear loading. (c)–(d), (e)–(f) and (g)–(h) show the formation, part-by-part movement and decomposition of the stair-rod dislocation, respectively. The magenta and green dislocation lines represent the stair-rod and Shockley dislocations, respectively. The atomic color represents structural type: green for FCC, red for HCP, blue for BCC and white for others.

adjacent Shockley dislocations through the following reaction (Fig. 4 (g)), even though it is energetically unfavorable:

$$\frac{1}{6}[\bar{1} 1 0] + \frac{1}{6}[\bar{1} 1 2] + 2 \times \frac{1}{6}[1 \bar{1} 2] \rightarrow \frac{1}{6}[\bar{2} 1 1] + \frac{1}{6}[2 \bar{1} 1] + \frac{1}{6}[1 1 2] + \frac{1}{6}[\bar{1} \bar{1} 2]. \quad (3)$$

This reaction allows GB migration to resume (Fig. 4(h)). The transient formation of stair-rod dislocations reduces the normal velocity $v_{\perp}$, resulting in a slightly higher β value than that predicted for the ideal $\langle 1 0 0 \rangle$ mode.

A similar migration driven by the dislocation glide was also found in the $\Sigma 243(1 1 22)$ GB (See Fig. S6). Note that the $\Sigma 243(1 1 22)$ GB is composed of the same type of $1/6 \langle 1 1 2 \rangle$ dislocations and exhibits similar collective glide behavior. However, the lower dislocation density induces the slower motion in this GB. It is worth noting that the previous study on the asymmetric tilt GBs also observed the dislocation interaction, including dislocation reaction and avoidance [40]. However, the mechanism observed here is distinct: although dislocation reactions and decompositions occur in the $\Sigma 163(1 1 18)$ and $\Sigma 243(1 1 22)$ GBs, each dislocation glide along a well-defined slip with minimal mutual interference.

Furthermore, this collective glide of multiple dislocations leads to distortion of the second invariant plane. In the ideal $\langle 1 0 0 \rangle$ mode, the $\{1 1 0\}$ plane serves as the second invariant plane, as exemplified by the $\Sigma 3(1 1 2)$ GB shown in Fig. S7, where the shear coupling is dominated by dislocations with Burgers vector $[0 0 1]$ [4,72]. However, as shown in Figs. S5(a) and (b) and S6(g) and (h), step-like feature emerges on the $\{1 1 0\}$ planes in the upper grain. This deviation is attributed to the meshed atomic movement caused by the two intersecting slip systems (See Fig. S8), which induces local distortions of the invariant plane.

For the high-angle $\Sigma 3(1 1 2)$ GB, shear-coupled migration can be described by the disconnection [68,69]. As shown in Fig. 5(a),

disconnections nucleate via significant displacements of atomic columns and subsequently propagate along the GB. The local atomic structure

(Fig. 5(b)) indicates that disconnection nucleation occurs through atomic columns (marked by orange arrows) filling the cavities of the lower E SUs. Analysis of the dichromatic pattern (Fig. 5(c)) reveals that the disconnection is characterized by a Burgers vector $\vec{b} = 1/3[1 \bar{1} \bar{1}]$ and a step height $h = 1/6|[1 \bar{1} 2]|$, yielding a coupling factor of $|\vec{b}|/h = -1.414$, consistent with the prediction of $\langle 1 0 0 \rangle$ mode.

It is noteworthy that, as a representative GB system, the shear response of the $\Sigma 3(1 1 2)$ GB has been extensively investigated [73,74]. Its behavior is strongly influenced by the stacking fault energy (SFE) of the metals. GB dissociation is typically observed in low-SFE metals, whereas sliding and migration dominates in high-SFE systems. In Pt, which has a relatively high SFE, the $\Sigma 3(1 1 2)$ GB shows shear-coupled migration behavior. Nevertheless, the disconnection characteristics and coupling mode differ from those in Al, where the size of Burgers vector is twice the $\{1 1 1\}$ interplanar distance, leading to a coupling factor approximately twice than that of Pt [73].

3.4. $\langle 1 1 0 \rangle$ mode

As shown in Fig. 3, the $\Sigma 33(4 4 1)$ and $\Sigma 51(5 5 1)$ GBs exhibit shear-coupling migration following the $\langle 1 1 0 \rangle$ coupling mode, with the coupling factor given by $\beta_{\langle 1 1 0 \rangle} = 2 \tan(\pi/2 - \theta/2)$. Both GBs consist of the same type of dislocation, therefore display similar migration mechanism (See Figs. 6 and S9-S11). Taking the $\Sigma 51(5 5 1)$ GB as an example (Fig. 6), migration proceeds through the glide of the $1/2 \langle 1 1 0 \rangle$ dislocations along $\{1 0 0\}$ planes, resulting in the $\{1 0 0\}$ plane serving as the second invariant plane (Fig. S9). Note that several works

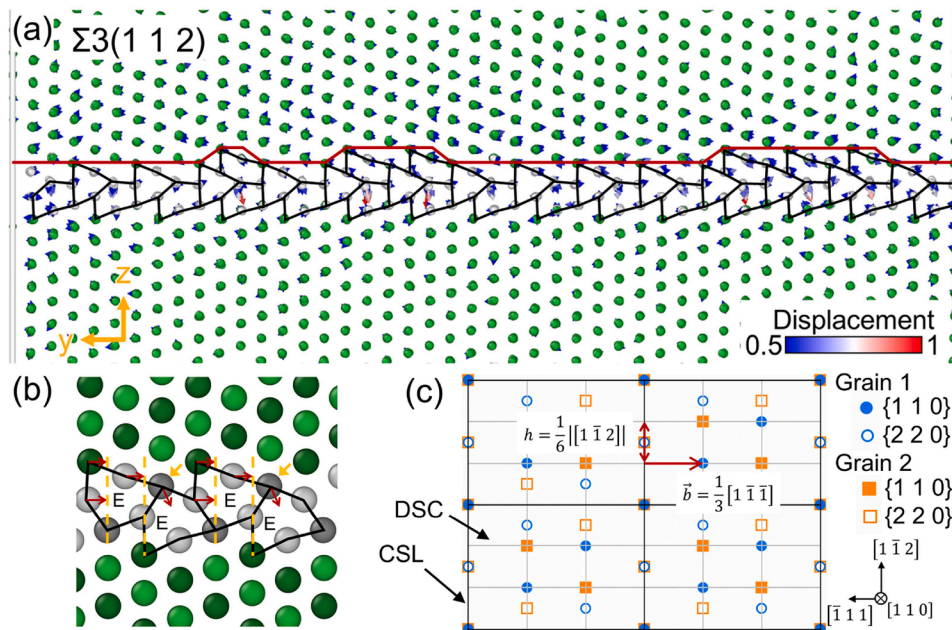


Fig. 5. (a) The atomic configuration of $\Sigma 3(1 1 2)$ GB with nucleated disconnections marked by the red line. The black lines show the deformed E SUs on the interface. The arrows represent atomic displacement colored by its magnitude. (b) The deformation process of E SUs on the GB plane. (c) The dichromatic pattern containing step height and Burgers vector of the disconnection. The CSL and displacement shift complete (DSC) lattice are shown with black and gray meshes, respectively.

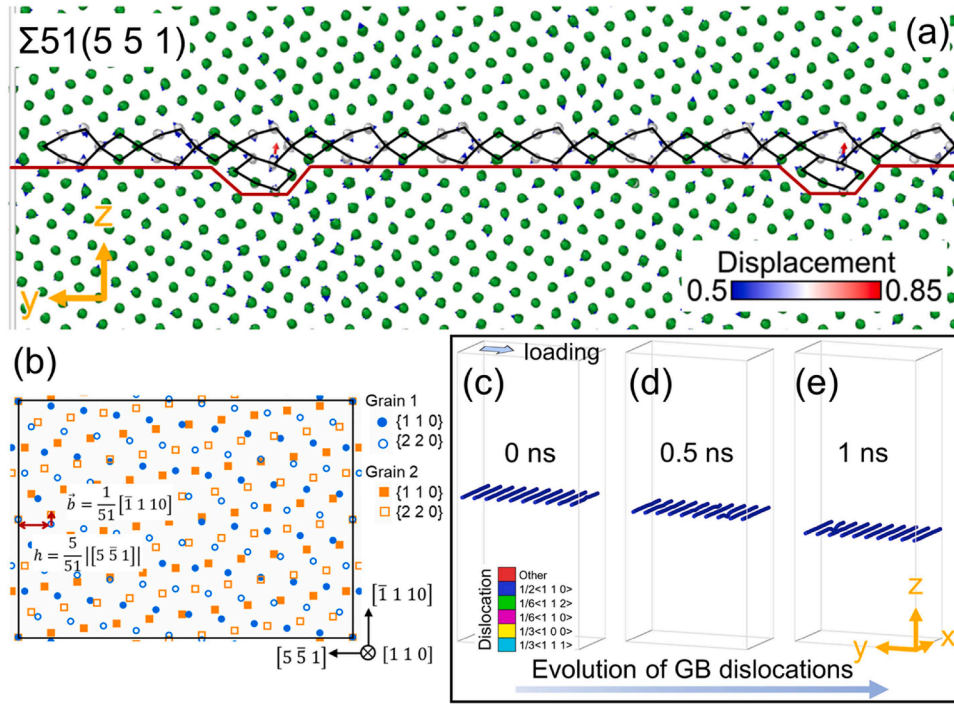


Fig. 6. (a) The $\Sigma 51(5\ 5\ 1)$ GB configuration with disconnections marked by the red line. The black lines show the deformed A' and E' SUs on the interface. The arrows represent atomic displacement colored by its magnitude. (b) The dichromatic pattern containing step height and Burgers vector of the disconnection. The black square represents the CSL. The DSC lattice is not shown due to its ultrasmall size. (c)–(e) show the evolution of GB dislocations under shear loading.

reported that $1/2\langle 1\ 1\ 0\rangle$ dislocation in FCC metals typically glide along $\{1\ 1\ 1\}$ plane [75,76]. This discrepancy may arise from a relatively lower slip barrier on $\{1\ 0\ 0\}$ plane for $1/2\langle 1\ 1\ 0\rangle$ dislocation in Pt. Moreover, the observed shear-coupled migration can also be interpreted using a disconnection-mediated process. As illustrated in Fig. 6(a), the movement of atom columns into the cavities in E' SUs generates a disconnection with a Burgers vector $\vec{b}$ of $1/51[1\ \bar{1}\ 10]$ and a step height h of $5/51||[5\ \bar{5}\ 1]||$, yielding a coupling factor β of $|\vec{b}|/h = 0.280$, consistent with the result obtained from MD simulations.

3.5. $\langle 1\ 1\ 2\rangle$ mode

As shown in Fig. 3, there are 4 GBs exhibiting the $\langle 1\ 1\ 2\rangle$ mode in

shear-coupled migration, including $\Sigma 11(1\ 1\ 3)$, $\Sigma 17(3\ 3\ 4)$, $\Sigma 57(4\ 4\ 5)$ and $\Sigma 3(1\ 1\ 1)$ GBs. The corresponding coupling factor is given by $\beta_{(1\ 1\ 2)} = -2\tan(\theta/2 - 70.53^\circ/2)$, where the 70.53° is the tilt angle of $\Sigma 3(1\ 1\ 2)$ GB. All of these GBs share $\{1\ 1\ 1\}$ plane as the second invariant plane (See Figs. S12–S15). For $\Sigma 11(1\ 1\ 3)$ and $\Sigma 3(1\ 1\ 1)$ GBs, shear-coupled migration can be well described by the disconnection (See Fig. S16). Specifically, disconnections with $\vec{b} = 1/22[3\ \bar{3}\ \bar{2}]$, $h = 2/11||[1\ \bar{1}\ 3]||$ are identified for the $\Sigma 11(1\ 1\ 3)$, while those for the $\Sigma 3(1\ 1\ 1)$ GBs are characterized by $\vec{b} = 1/6[1\ \bar{1}\ 2]$, $h = 1/3||[1\ \bar{1}\ 1]||$. In the $\Sigma 11(1\ 1\ 3)$ GB, the atom columns with C SUs move along the tilt axis, leading to the formation of a double-layer disconnection similar to that of Au bicrystal [30]. For $\Sigma 3(1\ 1\ 1)$ twin boundary, the relative slipping on the twin plane induces the disconnection nucleation.

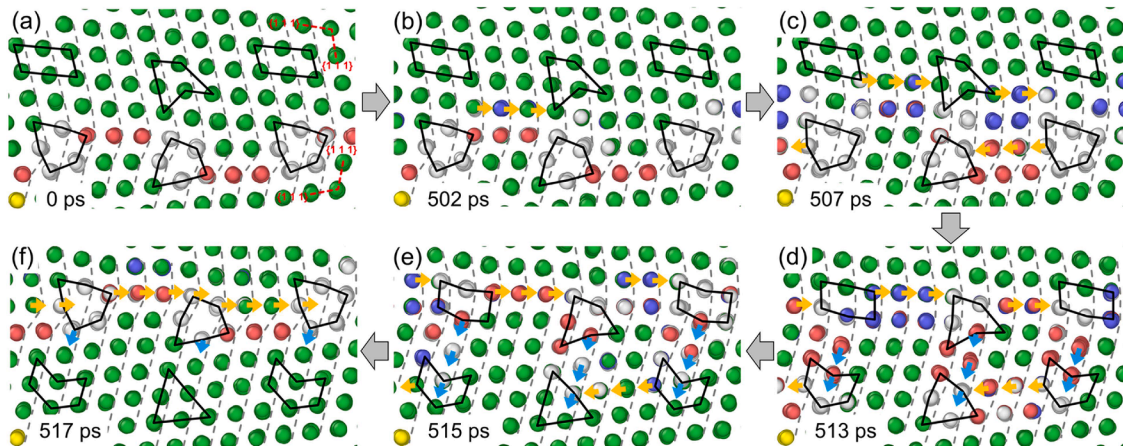


Fig. 7. The evolution of atomic structure during $\Sigma 57(4\ 4\ 5)$ GB migration. The atomic color represents structural type with green for FCC, red for HCP, blue for BCC and white for others. The structural evolutions of E' SUs are highlighted by black lines. The orientations of $\{1\ 1\ 1\}$ planes in the two grains are indicated by red dashed lines in (a). The orange and cyan arrows indicate the typical movement of atomic columns on different directions. The gray dashed lines marked the corresponding planes of the two grains. The yellow atomic column is set as a reference point in the figures.

Compared to the $\Sigma 11(1\ 1\ 3)$ and $\Sigma 3(1\ 1\ 1)$ GBs, the shear-coupled migration of $\Sigma 17(3\ 3\ 4)$ and $\Sigma 57(4\ 4\ 5)$ GBs is more complex. Taking the $\Sigma 57(4\ 4\ 5)$ GB as an example, the structural migration during migration reveals a multi-stage transformation process. Since D SU is the fundamental building block of twin boundary (See $\Sigma 3(1\ 1\ 1)$ in Fig. 2(g)), the $\Sigma 57(4\ 4\ 5)$ GB contains the segments of twin boundary composed of three consecutive D SUs (Fig. 2(f)). As shown in Figs. 7(a)–(c), shear deformation initializes through twin boundary sliding accompanied by the formation of a BCC intermediate phase, similar to that observed in the $\Sigma 3(1\ 1\ 1)$ GB. This sliding process is subsequently restrained by adjacent E SUs. With continued shear loading, the slipping of lattice plane along the $\{1\ 1\ 1\}$ plane (indicated by dashed lines) occurs, leading to the decomposition of existing E SUs and the formation of new E SUs in the upper grain (Figs. 7(d)–(f)). This twin-boundary-mediated structural transformation drives the overall shear-coupled migration of the $\Sigma 57(4\ 4\ 5)$ GB. Besides, due to the structural similarities, the $\Sigma 17(3\ 3\ 4)$ GB exhibits an analogous transformation pathway under shear loading, as shown in Fig. S17. This complicated migration behavior involves a mixed mechanism combining twin boundary slip and localized atomic shuffling, which is difficult to be described using the disconnection-mediated mechanism.

4. Discussion

In this work, we systematically investigated the deformation mechanisms underlying shear-coupled migration of $[1\ 1\ 0]$ symmetric tilt GBs in Pt, revealing a misorientation-dependent coupling behavior governed by distinct atomistic mechanism. Similar to $[0\ 0\ 1]$ symmetric tilt GBs and twin boundaries [4,10,77], each coupling mode corresponds to a specific second invariant plane which is found to be closely related to the tilt angle. For $[1\ 1\ 0]$ symmetric tilt GBs, three types of second invariant planes were identified: the $\{1\ 0\ 0\}$ plane for the $\langle 1\ 1\ 0 \rangle$ mode, the $\{1\ 1\ 0\}$ plane for the $\langle 1\ 0\ 0 \rangle$ mode and the $\{1\ 1\ 1\}$ plane for the $\langle 1\ 1\ 2 \rangle$ mode. Here we defined three angles, α , γ and δ , corresponding to the angles between pairs of $\{1\ 0\ 0\}$, $\{1\ 1\ 0\}$ and $\{1\ 1\ 1\}$ planes, respectively, as shown in Fig. 8(a). Meanwhile, a dimensionless parameter s used to

quantify the shear magnitude is defined as:

$$s = 2\tan(x/2) \quad (4)$$

where x represents α , γ or δ . A smaller value of s indicates a lower shear magnitude in the shear-coupled migration. The calculated variations of α , γ and δ and corresponding s values as a function of tilt angle are shown in Fig. 8(b). The results reveal a clear trend that the invariant plane associated with the smaller angle and lower s value is preferentially selected during shear-coupled migration. This behavior is analogous to the observations in deformation twinning and shear-coupled migration of $[0\ 0\ 1]$ symmetric tilt GBs [10,78].

It is noteworthy that $\Sigma 3(1\ 1\ 2)$ GB represents an exception to this general rule. Although $\langle 1\ 1\ 2 \rangle$ mode would suggest a $\{1\ 1\ 1\}$ invariant plane, the $\Sigma 3(1\ 1\ 2)$ GB instead follows the $\langle 1\ 0\ 0 \rangle$ mode with a $\{1\ 1\ 0\}$ invariant plane. It is seen that the angle between the two $\{1\ 1\ 1\}$ planes (δ) is 0° in the $\Sigma 3(1\ 1\ 2)$ GB, corresponding to a vanishing shear magnitude. This indicates that migration along the $\{1\ 1\ 1\}$ invariant planes would lead to decoupled normal GB motion without accompanying shear. However, such a process is prohibited due to the loading shear, and the GB instead adopts the suboptimal $\{1\ 1\ 0\}$ invariant plane. Notably, vertical migration of the $\Sigma 3(1\ 1\ 2)$ GB was indeed observed previously under different shear loading conditions [73].

Additionally, to further clarify the migration behavior near this extremal case, we have examined the GBs with the tilt angle close to that of the $\Sigma 3(1\ 1\ 2)$ GB. For the $\Sigma 353(8\ 8\ 15)$ GB (tilt angle = 74.05°), the structure is mainly composed of interconnected E SUs, similar to the $\Sigma 3(1\ 1\ 2)$ GB. This GB also exhibits a $\{1\ 1\ 0\}$ invariant plane during migration (Fig. S18). Detailed analysis of the structure evolution in both $\Sigma 353(8\ 8\ 15)$ and $\Sigma 3(1\ 1\ 2)$ GBs reveals similar transition pathways, resulting in identical invariant plane and coupling mode. Combined with the results for $\Sigma 17(3\ 3\ 4)$ and $\Sigma 57(4\ 4\ 5)$ GBs, these observations suggest that high- Σ GBs often contain the SUs inherited from neighbouring low- Σ GB with relatively low GB energy. As a result, the migration behaviors of high- Σ GBs are usually governed by the transformations of these embedded SUs, leading to coupling mode consistent with those of the corresponding low- Σ GBs. Therefore, we conclude that

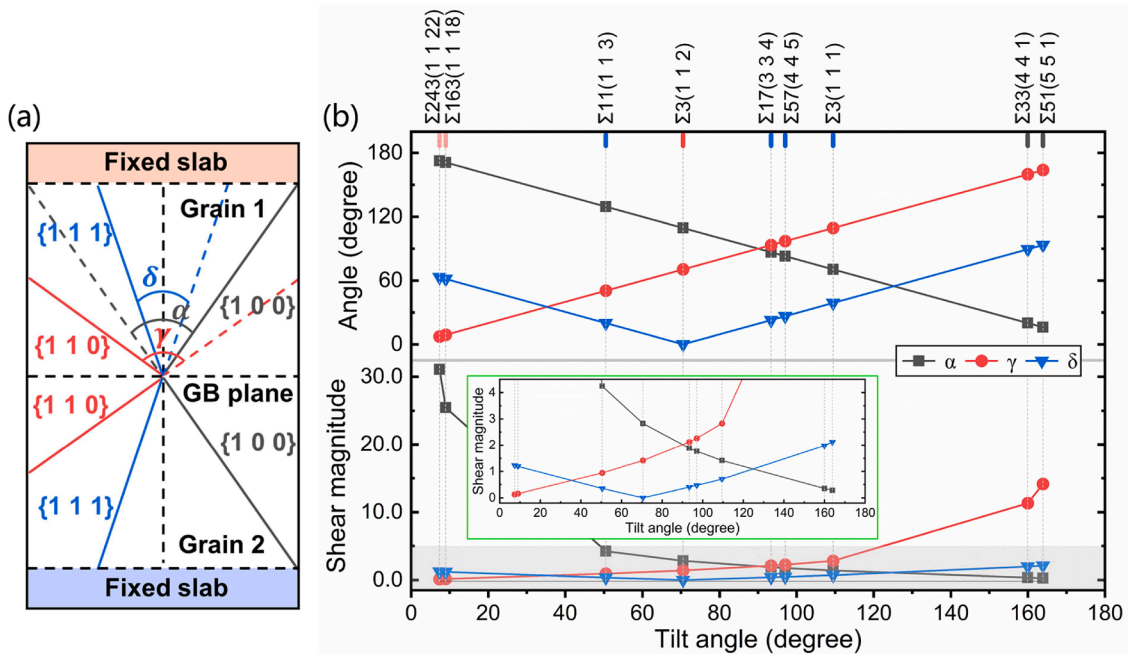


Fig. 8. (a) Schematic diagram of the angles of low-index plane pairs in GB model with $\{1\ 0\ 0\}$, $\{1\ 1\ 0\}$ and $\{1\ 1\ 1\}$ plane pairs marked by black (α), red (γ) and blue (δ), respectively. (b) Dependence of the angles of plane pairs (upper) and corresponding shear magnitude (lower) on the tilt angle. The inset in the lower plot shows the enlarged curves of the gray region. The color of top vertical lines denotes the second invariant plane of each GB with black for $\{1\ 0\ 0\}$, red for $\{1\ 1\ 0\}$ and blue for $\{1\ 1\ 1\}$.

shear-coupled migration is determined not only by geometric factors (e.g. tilt angle) but also by the specific atomistic structures (e.g. composed SUs).

We also considered the influence of GB metastability on the sheared-coupling migration. Previous work reported that, for $\Sigma 5(2\ 1\ 0)$ GB of Cu, the structure with split kites migrates in $\langle 1\ 0\ 0 \rangle$ mode, whereas the other one with filled kites migrates in $\langle 1\ 1\ 0 \rangle$ mode [39]. A similar result was observed in the $\Sigma 7(1\ 4\ 5)$ GB of Cu, where a domino phase exhibits a coupling factor different from a pearl phase [70]. Here we constructed two metastable $\Sigma 11(1\ 1\ 3)$ GBs and found that the structure metastability indeed affect the GB migration behavior. As shown in Fig. S19, during the migration, the ISFs nucleate and emit into the upper grain in one metastable model. In contrast, a phase transition to the stable complexion initially occurs in the other metastable phase. This result demonstrates the effect of metastability of GB complexion on the shear responses.

Moreover, we performed the simulations at elevated temperatures on the $\Sigma 11(1\ 1\ 3)$ GB to understand the temperature effect on the shear-coupled migration. The adopted temperature was increased with an increment of 300 K. The simulated results are shown in Fig. S20. We found that the coupling factor increases slightly with an increase of temperature around the ideal value for the $\langle 1\ 1\ 2 \rangle$ coupling mode. This may be attributed to thermal noise, i.e., the thermally induced uncompensated non-FCC atoms in two grains at higher temperatures, especially when GB moves away from the center. Furthermore, in the simulation of 1500 K, we observed a single-layer disconnections (Fig. S20(c)) which is different with the double-layer disconnection appeared in the other temperatures. In addition, the increased temperature can reduce the starting time of GB migration, i.e., the required shear stress for initial migration, as shown in Fig. S20(b).

5. Conclusion

We have systematically investigated shear-coupled migration of $[1\ 1\ 0]$ symmetric tilt GBs in Pt through large-scale MD simulations enabled by highly efficient machine learning potentials. Three coupling modes, $\langle 1\ 0\ 0 \rangle$, $\langle 1\ 1\ 0 \rangle$ and $\langle 1\ 1\ 2 \rangle$ mode, were identified, corresponding to second invariant planes of $\{1\ 1\ 0\}$, $\{1\ 0\ 0\}$ and $\{1\ 1\ 1\}$ plane, respectively. The novel mechanisms of GB migration were revealed. In particular, a collective intersecting dislocation glide was observed in low-angle $\Sigma 163(1\ 1\ 18)$ and $\Sigma 243(1\ 1\ 22)$ GBs with multiple dislocations, and a twin-boundary-mediated migration was found in the $\Sigma 17(3\ 3\ 4)$ and $\Sigma 57(4\ 4\ 5)$ GBs. Furthermore, we demonstrated that the coupling mode and associated second invariant plane depend not only on the tilt angle and shear magnitude, but also on the local atomic structure of the GB. This work provides a comprehensive understanding of shear-coupled migrations in $[1\ 1\ 0]$ symmetric tilt GBs of metals. The newly found migration mechanisms enrich to the current theoretical framework of GB migration. In addition, the machine learning based MD simulations scheme developed here offers a highly efficient and transferable strategy for investigating complex GB dynamics in a wide range of material systems.

CRedit authorship contribution statement

Zhenduo Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Junwen Duan:** Writing – original draft, Software, Methodology, Formal analysis. **Xuecheng Shao:** Writing – original draft, Methodology, Investigation. **Bo Gao:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.actamat.2026.122408](https://doi.org/10.1016/j.actamat.2026.122408).

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